Unit 2 Cluster Practice: Teacher Key

13 questions (7 multiple choice, 6 constructed response). Accept other scientifically accurate answers that show the same understanding.

Q	Answer	What earns credit / how to find it	Skill
1	**C → D → A → B**	Weathering, transport by gravity, deposition, compaction. Sandstone is sedimentary, so compaction is last.	Sequence a process
2	**(4)**	Gorge rocks (sandstone, shale, limestone) are sedimentary: burial, compaction, mineral precipitation.	Use a model's key
3	**(1)** slate, heat and pressure, deep in crust	Formed from shale buried during mountain building; splits into thin, flat sheets.	Classify a rock
4	Arrows: up from the food coloring, across the top, down the right side	Both parts: at least two arrows showing the loop **and** heated water is less dense, so it rises.	Develop a model
5	Basalt is mafic, not felsic	Name at least **two** minerals in basalt (p. 14): pyroxene, calcium-rich plagioclase, olivine.	Refute a claim
6	**(1)**	CO_2 (1.8 g/L) is denser than air (1.2 g/L), so it sinks into low places.	Use data
7	Geosphere → atmosphere	All three parts: from magma (geosphere); into the air (atmosphere, or biosphere if explained); leaks out of magma and seeps up through cracks and soil.	Track matter between systems
8	**(2)** B → D → A → C	Limestone buried, melts into magma, magma rises, CO_2 escapes.	Sequence a process
9	**(2)** less, more, increase	Caroga Lake is higher (450 m): water boils at a lower temperature, so more time is needed to raise the egg's temperature.	Read two graphs
10	**(2)**	Olivine absorbs hydrogen ions (less acidic, pH increases) and removes CO_2 from the ocean.	Interpret a model
11	Dissolves, then precipitates	Both steps: acidic groundwater **dissolves** calcite from limestone; in the cave CO_2 escapes and calcite **precipitates** as the stalactite.	Describe a process
12	**(2)** C, D and F identify calcite	B (depth) and E (mass) can't identify a mineral. Softer than glass, 3 cleavages not at 90°, reacts with acid.	Select the evidence
13	Potassium feldspar; felsic; quartz	All three parts. Also accept sodium-rich plagioclase, biotite, muscovite or amphibole. Not olivine or pyroxene.	Use two reference tables